# Resolution Geometry: The Realization Atlas

## Nine Layers from Exact Gluing to Physical Anchors, with a Certified Registry

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Companion to *Resolution Geometry* (placed after the ladder)

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**Abstract**

The ladder L0–L131 states what Resolution Geometry (RG) proves. The atlas states how a concrete system is *charted*: how local views of the observable quotient are glued, which structure survives each step, and which certificate a realization carries at each layer.

There are nine layers:

A1 exact set-level gluing, task descent and conversion;

A2 regular structured descent;

A3 metric charts;

A4 strata, walls and specialization;

A5 Γ-equivariant charts and slices;

A6 global gluing (cocycles, twisted forms, Leray covers, monodromy);

A7 quantitative stability (approximate gluing, conditioning, finite-sample charts);

A8 physical anchors;

A9 a registry of realizations with per-layer statuses.

Layer A1 is the existing layer-1 package. It is rerun: 5 651 contracts and 88 482 binary task checks. Layers A2–A8 are new executable layers; each has a contract, a theorem (classical where imported), witnesses and an explicit boundary. The registry places ten realizations, from a finite law map to the eccentric binary of the full computation loop.

Statuses belong to queries, not to systems. A realization can be certified at A1 and still be twisted at A6, or certified at A3 and ill-conditioned at A7. The atlas therefore never issues a single global verdict.

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# 1 What an atlas is in RG

**Definition 1.1** (Atlas, layer, certificate)**.** A *chart* is a pair $(V, \varphi)$, where $V \subseteq Q$ is a subset of the observable quotient and $\varphi$ is a bijection onto a model set, carrying whatever structure the layer declares. An *atlas layer* fixes three things:

- the structure carried by charts;
- the compatibility condition on overlaps;
- the query it answers.

A *certificate* is the layer's answer for one realization. It is one of: CERTIFIED, OBSTRUCTED (with a witness), SINGULAR, TWISTED, CONDITIONED (with its constant), CORRESPONDENCE, or NOT TESTED.

The layers are ordered by the structure they add. Each layer presupposes the certificates of the layers it uses, never the reverse.

| Layer | Structure | Question | Rungs realized |
|---|---|---|---|
| A1 | sets and equality of laws | does a local task glue to $Q$? does a conversion exist? | L0–L5, L9, L12–L15 |
| A2 | submersion, smooth tensors | which tensors descend? where does constant rank fail? | L18–L20 |
| A3 | metric (Fisher) | is the chart metric natural, tensorial, chart-free in distance? | L28–L35, L59–L63 |
| A4 | strata and walls | which specializations exist; where do charts break? | L21–L27, L37, L122 |
| A5 | effective symmetry $\Gamma$ | are charts equivariant; where are slices needed? | L6–L11, L25 |
| A6 | cocycles and global classes | does a global object, section or trivialization exist? | L111, L119–L122 |
| A7 | error budgets | how stable is gluing; how wide is a finite-sample chart? | L16, L44–L48, L114 |
| A8 | units and anchors | when is a chart a physical chart? | L94–L109 |
| A9 | registry | which realization carries which certificate? | all |

## 2 A1: exact gluing, task descent, conversion (imported layer 1)

**Theorem 2.1** (Effective set-level gluing; L1.1)**.** *Take the local quotients $Q_i = U_i/\sim$, glued by equality of their observation laws. The glued set is canonically $Q$, and compatible local tasks glue uniquely.*

**Theorem 2.2** (Sharp descent from raw overlaps; L1.2)**.** *Every raw-compatible family of locally basic tasks glues to $Q$ iff every fibre graph is connected. Two points of a fibre are joined when they share a chart. A saturated cover is sufficient but not necessary.*

**Theorem 2.3** (Conversion; L1.4)**.** *A unique conversion $h$ with $G = h \circ F$ exists iff $\ker F \subseteq \ker G$; otherwise the result is a correspondence. Relational composition can strictly over-approximate: $F = H = \mathrm{id}$ through a constant $G$ gives the full square. Joint witnesses must therefore be retained.*

*Witnesses (rerun).* Exhaustive enumeration for $|X| \leq 4$ and covers with at most three members:

- 5 651 contracts, of which 4 216 descend and 1 435 are obstructed by a disconnected fibre;
- 88 482 binary task checks;
- the cocycle mutation (pairwise transitions are invertible, yet a triple cocycle fails);
- marginal-versus-joint obstructions, including three pairwise anticorrelated bits with no joint law;
- graph covers of cycles with odd holonomy, which glue to an object that has no global section.

## 3 A2: regular structured descent

**Imported Theorem 3.1** (Descent through a submersion)**.** *Let $F : U \to Q$ be a surjective submersion with connected fibres. Then:*

- *A smooth function descends iff it is constant on fibres, i.e. $Vf = 0$ for vertical $V$.*
- *A vector field descends iff it is $F$-related.*
- *A covariant tensor descends iff it is* basic*: horizontal ($\iota_V T = 0$) and invariant ($\mathcal{L}_V T = 0$).*

*Witness.* Take $F(x, y, z) = (x^2 + y^2, z)$.

- The rank is 2 off the axis and 1 on it, so the axis is a singular stratum, handed to A4.
- The function $e^{x^2+y^2} \sin z$ descends; $xz$ does not.
- The radial field projects to $2s\,\partial_s$; $\partial_x$ does not project.
- A pulled-back metric is basic. The Euclidean metric is invariant but not horizontal, and does not descend.

*Boundary.* Descent of functions alone does not give descent of tensors: horizontality is an extra condition. Where the rank is not constant the theorem is silent, and A4 takes over.

## 4 A3: metric charts

**Imported Theorem 4.1** (Natural chart metric)**.** *The Fisher form is a natural module (L28). Its radical is* $\ker DF$ *on regular strata, it descends to* $Q$*, it transforms as a tensor under chart changes, and it is monotone under Markov garbling (L35). It is selected among natural metrics under* H-mono *(Čencov).*

*Witness.* Consider the redundant Gaussian with $\mu = a + b$.

- The latent Fisher form has rank 2, and its radical is spanned by $(1, -1, 0) = \ker DF$.
- The descended chart metric is $\mathrm{diag}(1, 2)/\sigma^2$.
- The chart change $\sigma = e^t$ transforms it exactly as a tensor.
- The geodesic distance between $(0, 1)$ and $(1.5, 2)$ is 1.39981 in the $(\mu, \sigma)$ chart, 1.39991 in the $(\mu, \log\sigma)$ chart, and 1.39988 in closed form, $\sqrt{2}\,\mathrm{arccosh}(\cdot)$. Distances are chart-free.
- Binning lowers the Fisher information from 1 to 0.8737 (data processing).

*Boundary.* A chart metric is selected under H-mono or by the task (quantum, L39), not by descent alone (L29). It is dimensionless; physical length is A8.

## 5 A4: strata, walls, specialization

**Theorem 5.1** (Wall charts; L21–L27)**.** *Kernels are upper semicontinuous, so walls are where the rank drops. A specialization map* $Q_\varepsilon \to Q_0$ *exists iff* $K_\varepsilon \subseteq K_0$*. Distinguishability across a wall needs Hellinger continuity (L37). On codimension-2 collision strata, charts of the regular stratum carry forced monodromy (L122).*

*Witness.*

- **Fold** $x^2$**.** The rank is 1 off the wall and 0 on it, the quotient chart has a boundary, and the odd task $x$ does not descend.
- **Specialization family** $(x_1, \varepsilon x_2)$**.** The map $(x_1, \varepsilon x_2) \mapsto x_1$ exists (checked on 80 seeded points, 40 of them sharing $x_1$); the reverse map does not. The Cramér–Rao bound for $x_2$ is 1, $10^2$ and $10^4$ at $\varepsilon = 1$, 0.1 and 0.01.
- **Eigenvalue collision.** Around it, the eigenline returns with sign $-1$.

*Boundary.* The classification of families whose type changes across walls is open (L131).

## 6 A5: Γ-equivariant charts and slices

**Theorem 6.1** (Equivariant charts; L6–L11, L25)**.** *Gauge (Gau) acts trivially on* $Q$*, and* $\Gamma = \mathrm{Stab}/\mathrm{Gau}$ *acts faithfully. Natural modules are* $\Gamma$*-invariant. Charts can be chosen equivariant on orbit-type strata. At points with larger stabilizer, a slice chart is needed: the cone points of* $Q/\Gamma$*.*

*Witness.* An isotropic Gaussian location in $\mathbb{R}^2$ with a nuisance angle.

- The nuisance is gauge and acts trivially on $Q$.
- $\Gamma = O(2)$ acts faithfully, and the Fisher module is invariant.
- Polar charts are equivariant ($\varphi \mapsto \varphi + \theta$).
- The stabilizer dimension is 1 at the origin and 0 elsewhere, so the origin is a cone point of $Q/\Gamma$.

# 7 A6: global gluing

**Imported Theorem 7.1** (Twisted forms and covers)**.** *Objects that are locally isomorphic to a model $x$ are classified by $\check{H}^1(S; \mathrm{Aut}_x)$. For a discrete structure group over a connected base this is $\mathrm{Hom}(\pi_1, G)/G$. Čech computations are valid on Leray covers. Triple defects with an abelian band define a class in $\check{H}^2$, which is removable iff it vanishes (L119–L121).*

*Witness.*

- A transported frame on $S^1$ returns with sign $-1$, although every local invariant is constant.
- Brute-force gauge orbits on a 3-cycle give 2 forms for $\mathbb{Z}_2$ and 3 for $S_3$ (its conjugacy classes).
- On three $250^\circ$ arcs of $S^1$, componentwise Čech cohomology gives $H^1 = \mathbb{Z}_2$, whereas a naive nerve would give 0.

*Boundary.* An atlas can glue to a global object without admitting a global section or trivialization (A1 graph covers, Möbius). These are different queries with different certificates.

# 8 A7: quantitative stability

**Proposition 8.1** (Approximate gluing and cycle certification)**.** (i) *If every transition in a chain of charts errs by at most $\delta$, chart values drift by at most $k\delta$ after $k$ steps.*

(ii) *A cycle holonomy $h$ is certified nonzero only if $|h| > \sum_{cycle} \delta_e$.*

(iii) *Near a wall the decoder has Lipschitz constant $1/\varepsilon$.*

(iv) *A finite-sample chart is a confidence set, and its tasks are returned as intervals (L16).*

*Witness.*

- A chain of 12 charts with $\delta = 10^{-4}$ has worst drift $2.3 \times 10^{-4} \le k\delta$.
- The cycle bound is $3.9 \times 10^{-4}$: a holonomy of half the bound is not certified, one of three times the bound is.
- The Lipschitz constants are 1, 10 and 100 at $\varepsilon = 1$, 0.1 and 0.01.
- A Bernoulli chart with $n = 400$ and Hoeffding radius 0.068 covers the truth in 0.9975 of 4000 seeded trials. Its log-odds task is returned as the interval $(-1.197, -0.541)$.

# 9 A8: physical anchors

**Theorem 9.1** (Charts become physical only through anchors; L94–L109)**.** *RG charts are dimensionless and unit-free, so they lie in a dilation fibre. A counted channel together with SI definitions removes the fibre ($d = c\,n/2\Delta\nu$). A held-out anchor validates the result.*

*Witness (illustrative numbers).* Two laboratories whose units differ by a factor 7.3 have the same chart (ratio 2.5). A counted caesium channel fixes $d = 3335.3913\,\mathrm{m}$.

# 10 A9: the registry

| Realization | Certificates by layer |
|---|---|
| R1 finite saturated law map | A1 CERTIFIED |
| R2 rotation quotient $(x^2+y^2, z)$ | A1 CERTIFIED; A2 CERTIFIED off the axis; A4 SINGULAR STRATUM (axis) |
| R3 redundant Gaussian $(\mu = a+b, \sigma)$ | A1, A2, A3 CERTIFIED |
| R4 fold and specialization family | A4 CERTIFIED (specialization), WALL (fold); A7 CONDITIONED $(1/\varepsilon)$ |
| R5 isotropic location with nuisance | A3 CERTIFIED; A5 CERTIFIED, cone point at the origin |
| R6 Möbius second-order family | A2 CERTIFIED; A6 TWISTED (no global section) |
| R7 noisy chart network | A7 CERTIFIED within bounds |
| R8 radar-anchored distance | A8 CERTIFIED |
| R9 Schwarzschild exterior | A2 CERTIFIED ($r > 2m$); A4 WALL (horizon chart boundary); A6 interior is a PERMANENT FIBRE; A8 CERTIFIED (radar; only $GM$) |
| R10 eccentric binary (full computation loop) | A3 tensor decoders; A6 WEYL HIDDEN STATE (nonlocal); A7 forecast budgets |

R9 and R10 import their certificates from companion papers. The validator checks that every layer cited in the registry was executed.

## 11 What the atlas does not do

1. It does not discover cross-chart equality of laws; that equality is a supplied input (A1).
2. It does not classify families across walls (L131), and gives no uniformly stable inverse there.
3. It does not certify statistical charts beyond declared classes and bounds.
4. It does not derive physical dynamics; charts describe states, not generators (L128).
5. It issues no global verdict: certificates answer different queries.

## 12 Battery

`rg_atlas_full.py` (NumPy, SciPy, SymPy; seed 20261102) reruns the layer-1 package and its validator (standard library), then executes A2–A8 and validates the registry. It takes about 9 s, and two runs give identical results.

**Limits.** The witnesses are finite, symbolic or seeded; the imported theorems are cited, not reproved. The A8 numbers are illustrative.

## References


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